

Grade 7
Unit 6
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Solve each problem using proportional reasoning. Show your work.

For questions 1-2, use the following scenario:

A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar.

1. How many tablespoons (tbsp) of vinegar are needed to make $\frac{1}{2}$ cup of salad dressing? 1 cup = 16 tbsp

$$\frac{1 \text{ cup}}{16 \text{ tbsp}} = \frac{\frac{1}{2} \text{ cup}}{8 \text{ tbsp}}$$

oil : vinegar : total $\rightarrow 3 : 1 : 4$

$$\frac{\text{vinegar}}{\text{total}} \left| \begin{array}{c} \xrightarrow{\times 2} \\ \frac{1}{4} = \frac{x}{8} \\ \xrightarrow{\times 2} \end{array} \right.$$

$$x = 2 \text{ tbsp}$$

2. How many tablespoons of oil are needed to make $\frac{1}{2}$ cup of salad dressing?

$$\frac{\text{oil}}{\text{total}} \left| \begin{array}{c} \xrightarrow{\times 2} \\ \frac{3}{4} = \frac{x}{8} \\ \xrightarrow{\times 2} \end{array} \right.$$

$$x = 6 \text{ tbsp}$$

3. The retail price of grapes is \$2.17 per pound. If a customer paid \$8.68 for grapes, how many pounds did she buy?

$$\frac{\$}{16} \left| \begin{array}{c} \xrightarrow{\times 4} \\ \frac{2.17}{1} = \frac{8.68}{x} \\ \xrightarrow{\times 4} \end{array} \right.$$

$$x = 4 \text{ pounds}$$

4. Gregg and Dan are moving boxes. Gregg can move 7 boxes every half hour. Dan can move 10 boxes every half hour. How many hours will it take Gregg and Dan to move all 85 boxes?

$$\frac{\text{boxes}}{\text{hours}} \left| \begin{array}{c} \xrightarrow{\times 5} \\ \frac{17}{0.5} = \frac{85}{x} \\ \xrightarrow{\times 5} \end{array} \right.$$

$$x = 2.5 \text{ hours}$$

For questions 5-7, use the following scenario:

A recipe for sparkling orange punch calls for $\frac{1}{2}$ gallon of orange sherbet, 6 ounces of frozen orange juice concentrate, and 2 quarts of ginger ale. Mei uses 3 gallons of orange sherbet to make punch for her party.

5. How much ginger ale does Mei need to use for her punch?

$$\begin{array}{c|c} \frac{\text{sherbert}}{\text{ginger ale}} & \frac{\frac{1}{2} \text{ gal}}{2 \text{ qt}} = \frac{3 \text{ gal}}{x} \end{array} \quad x = 12 \text{ quarts}$$

$\xrightarrow{\times 6}$
 $\xleftarrow{\times 6}$

6. How much frozen orange juice concentrate does Mei need to use for her punch?

$$\begin{array}{c|c} \frac{\text{sherbert}}{\text{OJ conc.}} & \frac{\frac{1}{2} \text{ gal.}}{6 \text{ oz.}} = \frac{3 \text{ gal.}}{x} \end{array} \quad x = 36 \text{ ounces}$$

$\xrightarrow{\times 6}$
 $\xleftarrow{\times 6}$

7. By what factor did Mei scale the recipe up?

6 She made 6 times the original recipe.

For questions 8-10, use the following scenario:

A teacher is planning a field trip to a theme park. For every 6 students, the teacher must have 1 chaperone.

8. What is the ratio of students to total people on the field trip?

6 : 7

9. If the teacher purchased 105 total tickets, how many tickets were for students?

$$\begin{array}{c|c} \frac{\text{students}}{\text{total}} & \frac{6}{7} = \frac{x}{105} \end{array} \quad x = 90 \text{ student tickets}$$

$\xrightarrow{\times 15}$
 $\xleftarrow{\times 15}$

10. If the teacher purchased 63 total tickets, how many tickets were for chaperones

$$\begin{array}{c|c} \frac{\text{chaperones}}{\text{total}} & \frac{1}{7} = \frac{x}{63} \end{array} \quad x = 9 \text{ chaperone tickets}$$

$\xrightarrow{\times 9}$
 $\xleftarrow{\times 9}$